

# What are the types and levels of cache memory

---

Knowledge Graph Reconstruction Report

April 2026

# CHAPTER 1

## WHAT WAS FOUND

Cache memory plays a critical role in bridging the performance gap between the processor and main memory. It acts as a buffer, residing within the processor to improve data retrieval speeds compared to RAM, which is slower. The construction of cache memory typically involves SRAM, chosen for its low latency characteristics

[src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d]. The historical significance of cache memory is underscored by its implementation in systems like the IBM System/360 Model 85, one of the pioneering platforms to utilize this technology [src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d].

Cache memory is organized into three main levels: L1, L2, and L3, each progressively larger and slower than the last. The L1 cache, for instance, is integral for day-to-day processor operations and is often on-chip, as exemplified by the 8 KB L1 cache introduced in the Intel 486DX [src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d]. These multi-level caches are designed to provide a hierarchical approach to data storage that optimizes access times and energy efficiency within a system.

There are distinct types of cache mapping: direct mapping, associative mapping, and set associative mapping. Direct mapping is characterized by each block of main memory being mapped to exactly one cache line, offering simplicity but potential for higher conflict misses when multiple active data lines map onto the same cache line

[src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d]. Associative mapping, contrastingly, allows a block to be stored in any cache line, reducing conflict misses but increasing lookup complexity. Set associative mapping strikes a balance by dividing the cache into sets, where each block can enter any line within a set. This method reduces miss rates while maintaining manageable complexity levels, as sets comprise several cache lines or blocks [src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d].

Performance metrics such as the hit and miss ratios are pivotal in evaluating cache efficacy. The hit ratio is determined by dividing the total number of cache hits by the sum of hits and misses, with higher ratios indicating efficient cache utilization

[src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d]. The miss ratio, complementarily, is the inverse of the hit ratio. These ratios directly influence the cycles per instruction (CPI), which measures the average number of clock cycles per executed instruction, thus serving as indicators of overall system performance [src:723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d].

Understanding these facets of cache memory, from structural levels to mapping types and performance metrics, is essential for appreciating how modern computing systems achieve efficient data processing speeds.

## CACHE MEMORY

- is\_constructed\_from -> SRAM (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)
- resides\_in -> processor (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)
- is\_faster\_than -> RAM (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)
- is\_buffer\_between -> central processor and main memory (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)
- levels -> L1 cache (source: inferred:cache-levels)
- levels -> L2 cache (source: inferred:cache-levels)
- levels -> L3 cache (source: inferred:cache-levels)
- types -> direct mapping (source: inferred:cache-types)
- types -> associative mapping (source: inferred:cache-types)
- types -> set associative mapping (source: inferred:cache-types)

## IBM SYSTEM/360 MODEL 85

- is\_historical\_milestone\_for -> cache memory (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## INTEL 486DX

- introduced -> 8 KB L1 cache (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)
- introduced -> on-chip 8 KB L1 cache (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## HIT RATIO

- is\_calculated\_as -> Total number of hits divided by (Total number of hits + Total number of misses) (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## MISS RATIO

- is\_calculated\_as -> 1 - hit ratio (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## **CPI**

- is\_calculated\_as -> number of cycles needed to execute one instruction (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## **IPC**

- is\_calculated\_as -> 1 divided by number of instructions (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## **DIRECT MAPPING**

- types -> cache mapping (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## **ASSOCIATIVE MAPPING**

- types -> cache mapping (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## **SET ASSOCIATIVE MAPPING**

- types -> cache mapping (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

- is\_constructed\_as -> divided into sets, each consisting of cache lines or blocks (source: 723c5b0e-c90a-476f-8da8-f7b1ef0dbf3d)

## CHAPTER 2

### CONFIDENCE LEVELS

cache memory -> is\_constructed\_from -> SRAM | confidence=0.95 (high)

cache memory -> resides\_in -> processor | confidence=0.95 (high)

IBM System/360 Model 85 -> is\_historical\_milestone\_for -> cache memory | confidence=0.90 (high)

Intel 486DX -> introduced -> 8 KB L1 cache | confidence=0.90 (high)

cache memory -> is\_faster\_than -> RAM | confidence=0.90 (high)

cache memory -> is\_buffer\_between -> central processor and main memory | confidence=0.95 (high)

Intel 486DX -> introduced -> on-chip 8 KB L1 cache | confidence=0.95 (high)

hit ratio -> is\_calculated\_as -> Total number of hits divided by (Total number of hits + Total number of misses) | confidence=0.95 (high)

miss ratio -> is\_calculated\_as -> 1 - hit ratio | confidence=0.95 (high)

CPI -> is\_calculated\_as -> number of cycles needed to execute one instruction | confidence=0.90 (high)

IPC -> is\_calculated\_as -> 1 divided by number of instructions | confidence=0.90 (high)

direct mapping -> types -> cache mapping | confidence=0.90 (high)

associative mapping -> types -> cache mapping | confidence=0.90 (high)

set associative mapping -> types -> cache mapping | confidence=0.90 (high)

set associative mapping -> is\_constructed\_as -> divided into sets, each consisting of cache lines or blocks | confidence=0.90 (high)

cache memory -> levels -> L1 cache | confidence=0.80 (high)

cache memory -> levels -> L2 cache | confidence=0.80 (high)

cache memory -> levels -> L3 cache | confidence=0.80 (high)

cache memory -> types -> direct mapping | confidence=0.80 (high)

cache memory -> types -> associative mapping | confidence=0.80 (high)

cache memory -> types -> set associative mapping | confidence=0.80 (high)

## **CHAPTER 3**

### **CONFLICTS**

No conflicts detected.

## CHAPTER 4

### WHAT IS MISSING

- Node 'cache memory' missing attributes: size, speed
- Node 'SRAM' missing attributes: size, speed
- Node 'RAM' missing attributes: size, speed
- Node 'L1 cache' missing attributes: size, speed
- Node 'L2 cache' missing attributes: size, speed
- Node 'L3 cache' missing attributes: size, speed
- Have all primary cache levels (L1, L2, L3) been identified? -> Not confirmed in graph (but may appear in narrative)
- Are the functions of each cache type described? -> Not confirmed in graph (but may appear in narrative)

## **CHAPTER 5**

### **SOURCES USED**

- a06e149a - 10.11648.j.ajece.20230702.11.pdf
- 00bb6791 - COMPUTER ORGANIZATION NOTES.pdf
- 7d8d1f0d - DLD1.pdf
- 723c5b0e - LITERATURE\_SURVEY\_ON\_CACHE\_MEMORY\_ijariie9590.pdf
- 10dd6156 - recent advancements in DRAM nad flash memory.pdf